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Digital Assistants at Microsoft Research
(Courtesy: Microsoft Research)

many forms of complex, layered interaction between machines and humans.

People who visit Horvitz at his Redmond office are often pleasantly surprised by the practical results of the research. When you get to his office floor, a cute little robot would greet you, give you directions with proper hand movements, and let Horvitz's virtual assistant Monica know that you have arrived. Monica, an onscreen personality, greets you in her unmistakable British accent. In the background, she does a quick calculation about the costs and benefits of an unplanned interruption, taking into consideration Horvitz's current desktop activity, calendar and past behaviour. When she is convinced that a short interruption is okay, she lets you in. Monica is smarter than this; she can access Horvitz's online calendar, detect his presence in office, predict when he will finish a task, infer how busy he is, when he will return to office after a meeting and even when he will read and reply to emails, and even predict, quite well, when he will conclude a phone conversation based on her 'observation' of his past behaviour!

Machines that are part of this project have situational awareness and can take into account the physical aspects of an interaction including people's gait, gestures, etc, which helps understand the tone of an interaction better. The project integrates machine vision, natural language processing, machine learning, automated planning, speech recognition, acoustical analysis, sociolinguistics, etc, to help robots and computers better understand multi-party interaction. The team has tried to endow systems with short-term and long-term memory so they can use past

experiences also to decide how to react in a particular situation. In a Microsoft Research article, Horvitz comments that, "Interactions with the system, especially when we integrate new competencies, seem magical, with the magic emerging from a rich symphony of perception, reasoning and decision-making that's running all the time."

Help me, haathi

An elephant is one of the most intelligent creatures on our planet, and undoubtedly its trunk is one of nature's smartest creations. It features tens of thousands of muscles working in tandem to deftly break a small nut as beautifully as it can uproot a large tree! This versatility of the elephant's trunk has long inspired robotics scientists, and ultimately a few months ago, scientists of German engineering firm Festo managed to develop a bionic elephant trunk that can learn and work just like an elephant's.

The trunk is made of 3D-printed segments and is controlled by a system of pneumatic muscles. Instead of using regular precision control software, the team has used a method called 'goal babbling' or trial-and-error learning, inspired by how infants learn to use their muscles. When the robot works, it records the tiny pressure adjustments made in the pneumatic tubes operating the artificial muscles. It then creates a map that relays the trunk's exact position, to calibrate the pressure in each tube.

The trunk can be trained by manually manipulating it into required positions. While it might resist the movement at first, the trunk slowly yields and follows the movement. It then 'learns' it. So, the next time you start pushing it to the same position, it moves easily and naturally. They call this ability of the trunk to remember its past movements as muscle memory.

The company hopes that this muscle memory together with the trunk's dexterity will enable its use in industrial environments and for jobs like changing road lights, picking apples, etc.